

# Maximizing Solar Panel Output for a Fixed Area

In this project, our team **Solar Panel Team #3** designed a solution to identify the **optimal tilt angle** and **optimal aspect ratio** of a solar panel to deliver the **maximal energy output** given environmental constraints. Additionally this project maps the optimal energy output onto a surf plot to illustrate how energy output is dependent on tilt angle and aspect ratio.

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## MODEL 1: Maximizing Energy Output given fixed constraints

The purpose of this model was to provide a way of finding the Maximum Energy Output given fixed environmental constraints

### Objective 1: Implement the Optimization Problem

The [optimization problem](#) solves for the **Maximum Energy Output**.

```
% Step 1: Define the Optimization Problem (Solar_Max). Give it a description
("Description") of its purpose and solve for Maximum Value Output ("maximize")
    Solar_Max = optimproblem("Description","The Maximal Energy
Output","ObjectiveSense","maximize");
```

By defining the optimization variables **Tilt Angle** ( $\theta$ ) and **Aspect Ratio** ( $r$ ), the solver finds the highest energy output given the fixed constraints of Tilt Angle and Aspect Ratio.

In Model 1, the environmental **Constraints are Fixed**, based off a theoretical assumption given in the project description.

- Constraint for Tilt Angle:  $0^\circ \leq \theta \leq 90^\circ$
- Constraint for Aspect Ratio:  $0.5 \leq r \leq 4$

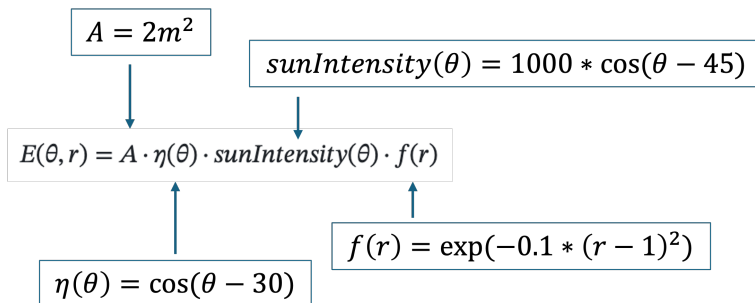
```
% Step 2: Define Optimization Variables and set Constraints

    % Constrain the lowest possible tilt angle to 0 degrees and the highest to 90
degrees
    theta = optimvar("theta","lowerbound",0, "UpperBound",90);

    % Constrain the lowest possible aspect ratio to 0.5 and highest to 4
    r = optimvar("r","lowerbound",0.5,"UpperBound", 4);
```

## Objective 2: Define the Objective Function & Solve for Optimal Tilt Angle and Aspect Ratio

The energy function is the objective function for this optimization problem. It calculates how much energy a solar panel produces based on its **Tilt Angle (theta)** and **Aspect Ratio (r)**



Note: The solver uses radian values instead of degrees. Multiplying inside the cosine function with  $\pi/180$  convert the theta values from degrees to radians.

```
% Step 1: Define "A" (the fixed panel area) as 2m^2
A = 2;

% Step 2: Define "nu" (the angle efficiency) in terms of the optimization variable
theta
nu = cos((theta - 30)*pi/180);

% Step 3: Define "fr" (shape efficiency factor) in terms of the optimization
variable r
fr = exp(-0.1*(r - 1)^2);

% Step 4: Define "Sun_Intensity" (variation in sunlight) in terms of the
optimization variable theta
Sun_Intensity = 1000*cos((theta - 45)*pi/180);

% Step 5: Define the Objective Function "Energy" (Total Energy Output) in terms of
"A", "nu", "Sun_Intensity" & "fr" which are in terms of "theta" and "r"
Energy = A.*nu.*Sun_Intensity.*fr;

% Step 6: Create the Objective Function (Solar_Max.Objective)
Solar_Max.Objective = Energy;

% Step 7: Set up the Initial Guess for "theta" and "r" (choose values within the
constraints)
initial_guess.theta = 45;
initial_guess.r = 2;

% Step 8: Use the solve function to find the optimal Aspect ratio & Tilt angle
which delivers the Maximum Energy Output
```

```
[ solution, maxE ] = solve(Solar_Max,initial_guess);
```

Solving problem using fmincon.

Local minimum found that satisfies the constraints.

Optimization completed because the objective function is non-decreasing in feasible directions, to within the value of the optimality tolerance, and constraints are satisfied to within the value of the constraint tolerance.

<stopping criteria details>

#### % Step 9: Store Results

```
Optimal_theta = solution.theta;
```

```
Optimal_ratio = solution.r;
```

```
Optimal_energy = maxE;
```

## Objective 3: Output the Results

**Plot  $E(\theta, r)$**  to visualize the energy as a surface plot using `meshgrid` and `surf`. Plot the optimal tilt angle and aspect ratio, and corresponding energy output onto the grid; the absolute maximum of the surface plot should align with the optimal point. Finally, print the optimal tilt angle and aspect ratio, and the Maximum energy output.

#### % Step 1: Create Grid Values for Tilt Angle and Aspect Ratio using meshgrid

```
% Set the lower bound to 0, the upper bound to 90 and the steps to 1
```

```
theta_values = 0:1:90;
```

```
% Set the lower bound to 0.5, the upper bound to 4 and the steps to 0.1
```

```
r_values = 0.5:0.1:4;
```

```
% Create the meshgrid
```

```
[theta_grid, r_grid] = meshgrid(theta_values,r_values);
```

#### % Step 2: Generate energy grid values that are dependent on the "theta\_grid" and "r\_grid"

```
% Note: It's identical to "Energy" the objective function
```

```
energy_grid = 2.*(cos((theta_grid-30).*pi/180)).*(1000.*cos((theta_grid-45).*pi/180)).*(exp(-0.1.*(r_grid-1).^2));
```

#### % Step 3: Use figure function to position the size and location of the plot

```
figure("Position",[100,100, 1000, 700]);
```

#### % Step 4: Create the Surface Plot using the surf function

```
surf(theta_grid,r_grid,energy_grid);
```

```
% Add shading that illustrate the gradual change in energy output using
```

```
% shading and colorbar functions
```

```
shading interp;
```

```
colorbar;
```

```
% Set Axis limits for the plot (equal to or greater then constraints)
```

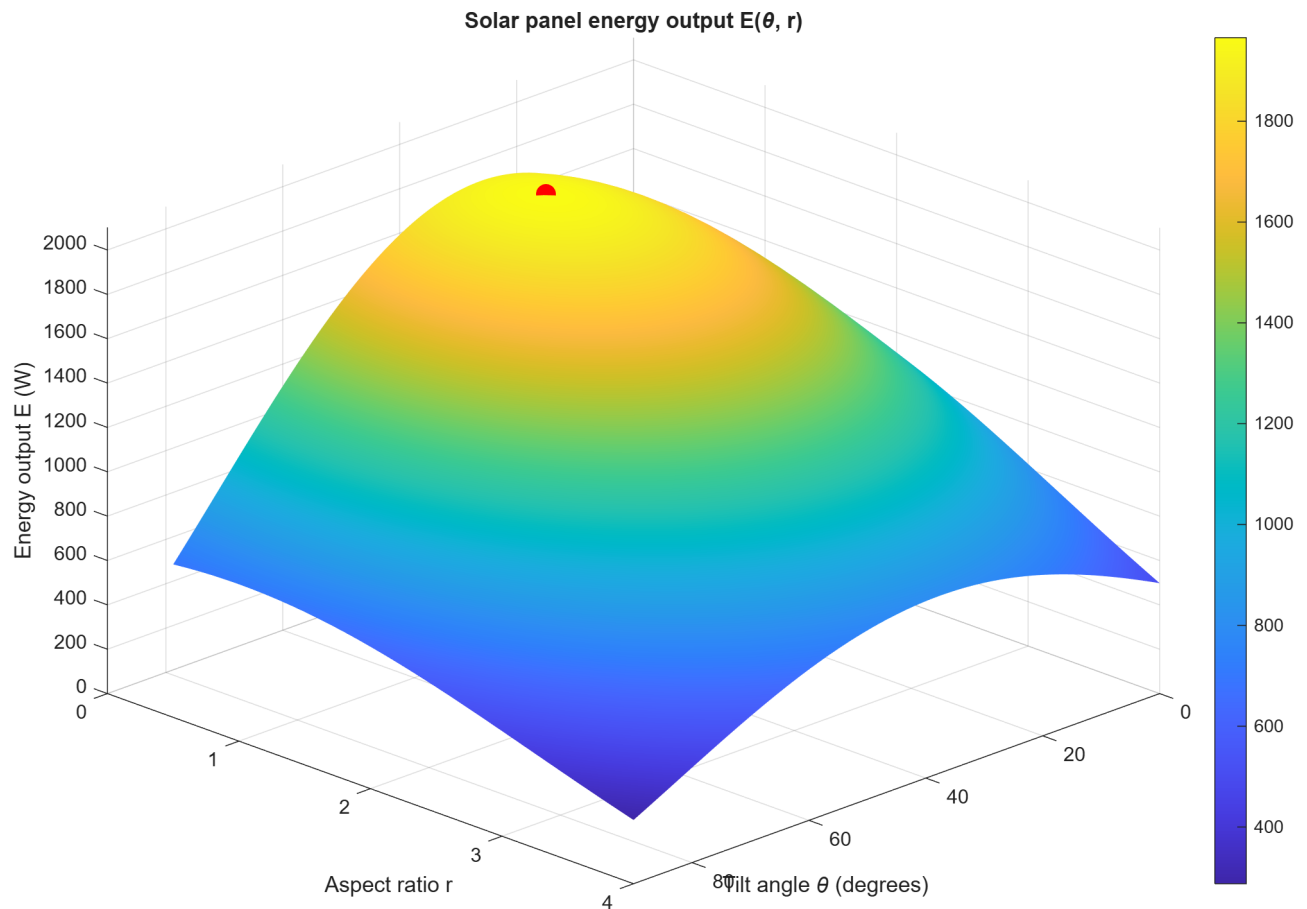
```

    xlim([0 90]);
    ylim([0 4]);
    zlim([0 2100]);
    % The view function can be changed to gain a better camera line of sight of the
    surface plot
    view(135, 30);

% Step 5: Label the plots
xlabel("Tilt angle \theta (degrees)");
ylabel("Aspect ratio r");
zlabel("Energy output E (W)");
title("Solar panel energy output E(\theta, r)");

% Step 6: Plot the optimal point on the surface plot
hold on;
plot3(Optimal_theta, Optimal_ratio, Optimal_energy, "r.", "MarkerSize", 30);
hold off;

```



```

% Step 7: Print Results

```

```
fprintf("Optimal Tilt Angle : %.2f degrees\n", solution.theta);
```

Optimal Tilt Angle : 37.50 degrees

```
fprintf("Optimal aspect ratio: %.2f\n", solution.r);
```

Optimal aspect ratio: 1.00

```
fprintf("Maximum energy : %.2f W/m^2\n", maxE);
```

Maximum energy : 1965.93 W/m<sup>2</sup>